

INTERNAL – SOLUTION SPEC · V1.0

Emfuleni Water & *Billing Platform*

Solving the revenue-side crisis through accurate metering, billing reconciliation, and resident payment access.

VERSION	ISSUED	AUTHOR	STATUS	SOURCE
1.0	MAY 2026	PRIMESERVE	INTERNAL ONLY	EMFULENI IDP 2026/27

What we cover today

Seven sections — context, problem, opportunity, solution, dependencies, market, and next steps.

01

Context

Who Emfuleni is, the financial reality, and why July 2026 is the window.

03

Opportunity

What the IDP says about billing — in their own words — and the ROI case.

05

Dependencies

Six things we must confirm before writing a line of code.

07

Next steps

What business and developers each own in the next four weeks.

02

The problem

Water infrastructure, billing breakdown, and the vicious cycle that links them.

04

The solution

Platform modules, technical surfaces, AI use cases, and phased delivery.

06

Go-to-market

Entry strategy, procurement pathway, and revenue model options.

SEVEN SECTIONS.
SIXTEEN SLIDES.
≈25 MINUTES.

Who is Emfuleni — *and why now?*

EMFULENI LOCAL MUNICIPALITY

Covers Vereeniging, Vanderbijlpark, Sebokeng, Evaton, Sharpeville, Boipatong, and Bophelong — the population heart of the Sedibeng District.

79% of the district's residents live here on 23% of its land. The municipality is in formal financial distress — operating under a Financial Recovery Plan (Dec 2024) with 7 of 8 consecutive qualified audit opinions.

POPULATION (CENSUS 2022) 945,650	UNEMPLOYMENT RATE 40%
MUNICIPAL BUDGET 2025/26 R9.6_{bn}	REVENUE COLLECTION RATE ~75%

WHY NOW?

The Vaal Central Water Utility (VCWU) — a Special Purpose Vehicle — takes full operational control of water services on **1 July 2026**. This creates an urgent, unsolved reconciliation gap between bulk water supply and retail billing — precisely the gap our platform is designed to close. The window is now.

Three interconnected failures — *one systemic crisis.*



Why the problem is self-reinforcing — *and where we break it.*

1. Infrastructure fails

Aged pipes burst. Meters break. Pump stations offline. Illegal connections tap supply. 56% never reaches a billing point.

2. Water goes unbilled

Lost or unmetered water generates no invoice. The municipality pays to treat and pump it — and receives nothing in return.

3. *Inaccurate bills* → *non-payment*

Estimated bills are disputed. Residents with unreliable service refuse to pay. Collection rate sits at ~75%, below the 80%+ target.

4. Revenue shortfall → no maintenance

Funding gaps mean no ring-fenced maintenance budget. The 70% Metsi-a-Lekoa vacancy rate means even funded repairs cannot execute.

5. Infrastructure worsens

More pipes fail. More meters break. More losses accumulate. The cycle resets at a worse baseline than before.

WHERE WE INTERVENE

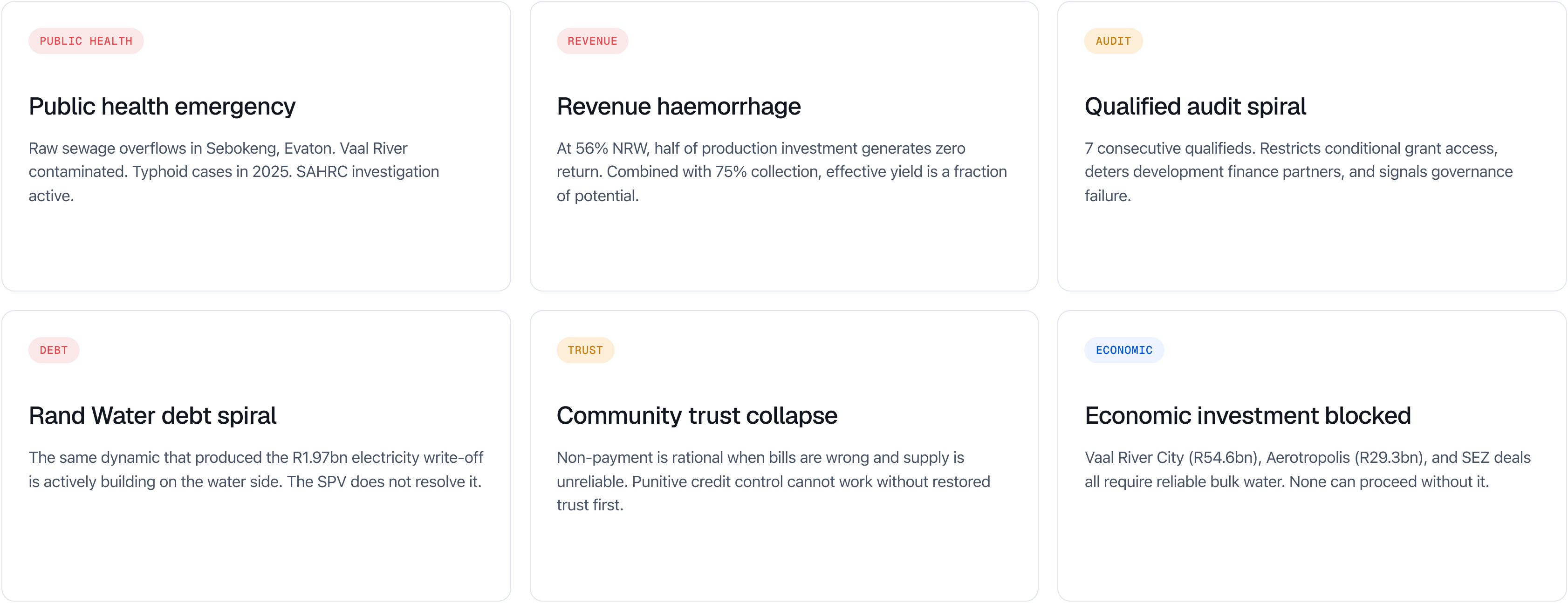
Steps 2 & 3

We fix the link between meter data and billing, and between billing and payment.

Accurate meters → accurate bills → restored trust → higher collection → maintenance budget.

We do not fix the pipes.
That is the SPV's job.

What the crisis is causing *downstream*.



1 JULY 2026

What changes — *and what does not.*

VCWU / SPV

Vaal Central Water Utility takes over

- Water supply systems & distribution network
- Sewer networks & reticulation
- Wastewater treatment works (Leeuwkuil, Rietspruit, Sebokeng)
- Operations, maintenance & future capital upgrades
- Rand Water as implementing agent (Section 63)

OUR CUSTOMER

Finance Cluster retains

- Customer accounts & billing system
- Revenue collection & credit control
- Indigent register management
- Valuation Roll reconciliation mandate
- Dispute resolution & account queries

CRITICAL INSIGHT

The SPV makes the reconciliation problem *more* complex, not less. After 1 July 2026, Emfuleni Finance must reconcile VCWU bulk supply against retail billing across an entity boundary. The market for a reconciliation platform grows on SPV day — it does not shrink.

Their diagnosis — *in their own words.*

“Billing system challenges have affected consumer confidence and revenue collection.”

IDP SWOT — LISTED AS A TOP WEAKNESS

“The billing system must be updated to ensure all consumers are billed accurately.”

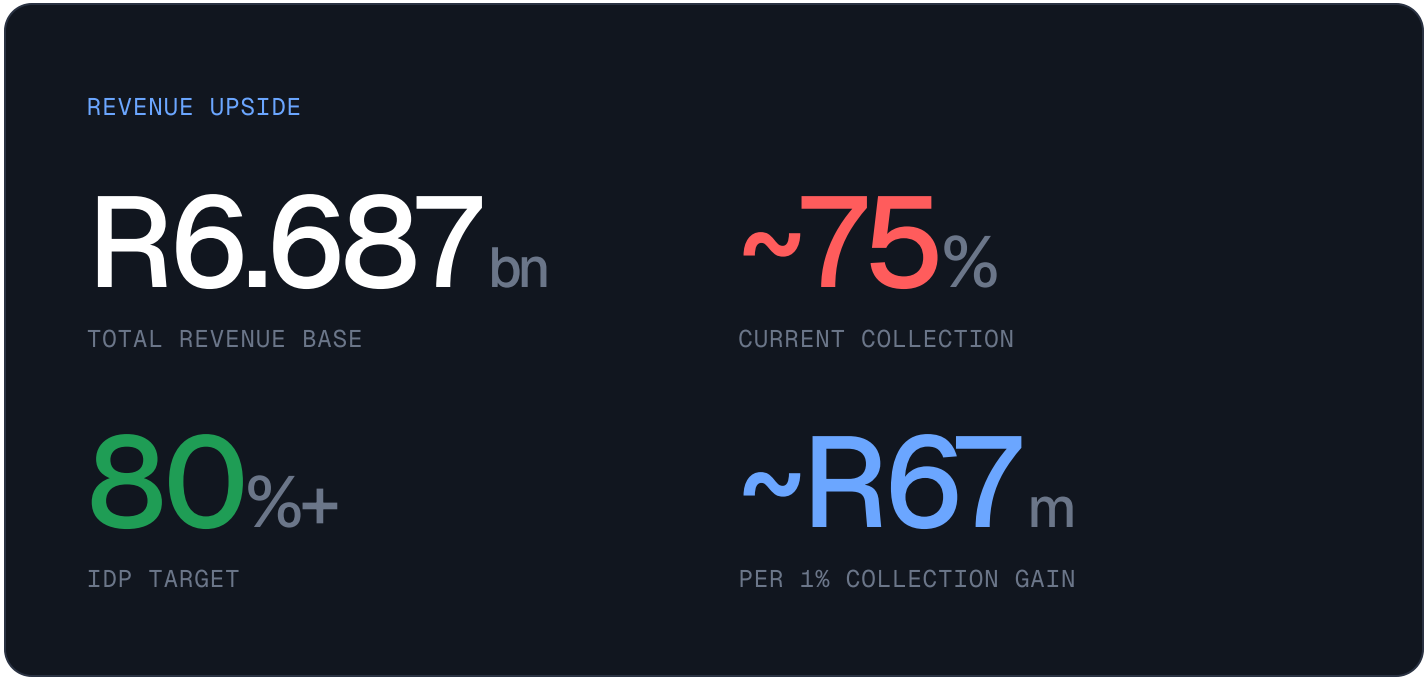
REVENUE MANAGEMENT CHAPTER — FORMAL MANDATE

“Digital transformation: improved municipal systems for billing, reporting, and service delivery.”

IDP SWOT — STRATEGIC OPPORTUNITY

“Addressing the low collection rate remains our foremost priority.”

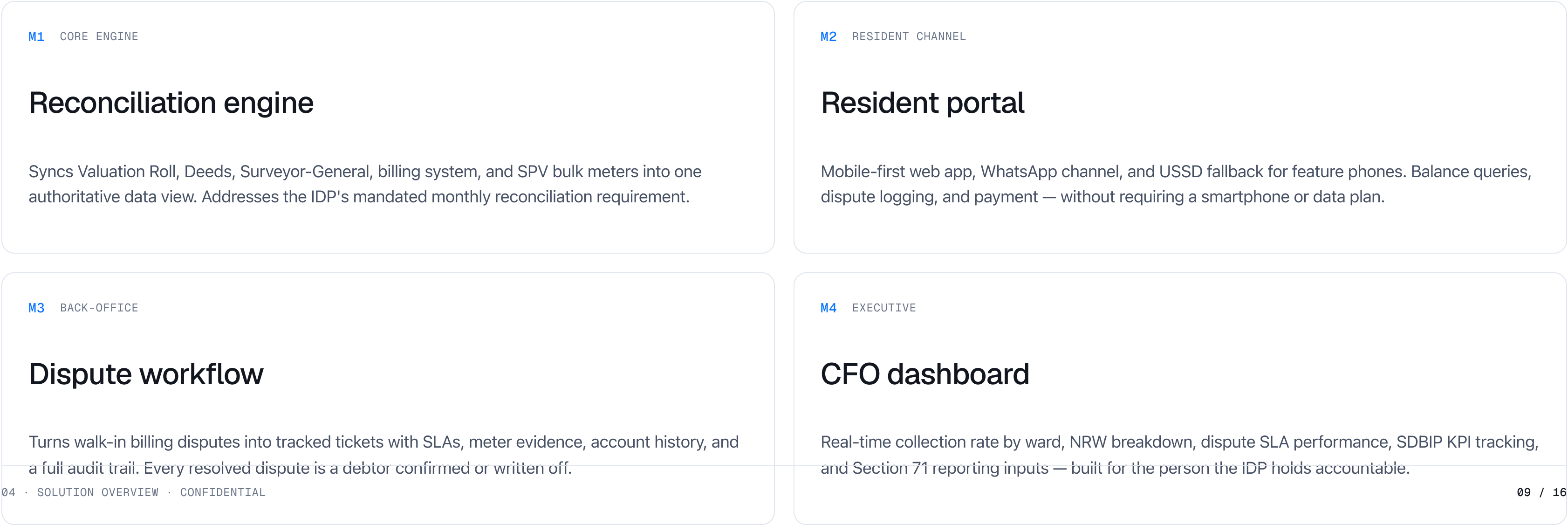
EXECUTIVE MAYOR'S FOREWORD



These are not our words.
They are Emfuleni's own diagnosis.

What we are *building*.

A multi-surface billing, reconciliation, and payment platform for Emfuleni's Finance Cluster — connecting meter data to accurate invoices, and invoices to resident payments.



Platform surfaces — *who uses what, and how.*

RESIDENT-FACING

Mobile web app

Android-first. Bill view, dispute logging, payment. Works on data; no app store required.

WhatsApp Business API

Balance queries, alerts, dispute intake — highest penetration channel in SA townships.

USSD *xxx#

Feature-phone access for Sebokeng/Evaton residents without data plans.

SMS notifications

Bill ready, payment confirmed, meter reading reminders.

MUNICIPAL / SPV-FACING

Back-office web app

Clerk dispute queue, meter fault tracking, account search — desktop browser, keyboard-first.

Reconciliation dashboard

Valuation Roll ↔ Deeds ↔ SG ↔ Billing ↔ VCWU bulk meter matching — the IDP's mandated reconciliation.

Field mobile app

Android, offline-capable. AI photo meter capture for field crews; syncs when connectivity returns.

EXECUTIVE / COUNCIL

CFO live dashboard

Collection rate by ward, age analysis, dispute SLAs, NRW breakdown.

Section 71 reporting

Automated monthly financial performance inputs for Council compliance.

SHARED PLATFORM

All surfaces read from the same reconciled data layer — no duplicate truth.

Technology choices — *and where AI earns its keep.*

LAYER	CHOICE	RATIONALE
Backend	Python / FastAPI or Node / TypeScript	Transactional billing; strong consistency & audit logs.
Database	PostgreSQL + DuckDB / ClickHouse	OLTP for billing + columnar analytics for reconciliation.
Hosting	AWS af-south-1 or Azure SA North	POPIA data residency — SA resident data stays in SA.
Auth	SMS OTP (residents) · SSO / MFA (back-office)	Residents: account + ID + OTP. Staff: AD/Azure AD + MFA.
Notifications	SMSPortal + WhatsApp Business API	Abstracted gateway — not coupled to a single channel.
USSD	Clickatell / Flickswitch aggregator	Stateless menu machine; our backend, not our infra.
Payments	Ozow (instant EFT), PayShap, Easypay	Ozow for digital; Easypay for cash at Shoprite / Pep.
Meter photo read	Vision AI (Anyline / custom model)	Far better digit accuracy than Tesseract on outdoor meters.

WHERE AI EARNS ITS KEEP

- ✓ Meter photo → digit extraction
- ✓ Anomaly detection (10× spike = leak or meter fault)
- ✓ Dispute classification & routing
- ✓ Indigent document parsing
- ✓ WhatsApp conversational queries

NEVER IN

- ✗ Tariff calculation
- ✗ Ledger operations
- ✗ Compliance reporting

Those must be deterministic and auditable.

Build sequence — *wedge first, vision second.*

Phase 1

MONTHS 0-6

Disputes + visibility wedge

1. Dispute / query workflow — clerk queue, SLAs, evidence trail, audit log.

2. Resident WhatsApp + responsive web: balance view and dispute logging.

3. Read-only integration to existing billing system via nightly file exports.

4. AI dispute classification and indigent document parsing.

DELIVERABLE

Measurable dispute backlog reduction and confirmed-debtor collection.

Phase 2

MONTHS 6-18

Reconciliation engine + payments

1. Valuation Roll ↔ Deeds ↔ SG ↔ Billing reconciliation engine (IDP mandate).

2. Payment processing: Ozow, PayShap, Easypay — transaction fee revenue model.

3. Field mobile app with AI meter photo reading for field crews.

4. VCWU bulk supply reconciliation layer (post-SPV July 2026).

DELIVERABLE

Live reconciliation dashboard and payment gateway active.

Phase 3

MONTHS 18+

Scale & replicate

1. USSD channel live — feature-phone township access.

2. Smart meter data ingestion (electricity now, water progressively).

3. Multi-tenant rollout: second and third municipality.

4. CFO SDBIP / Section 71 automated reporting module.

DELIVERABLE

Reference customer and replicable multi-tenant platform.

Rule: build only what the current phase requires. The municipality cannot absorb large-scale change quickly — ship measurable results before the next phase begins.

What we must confirm — *before writing a line of code.*

D1HIGH Existing billing system vendor Changes the entire integration architecture — Solar, Munsoft, Sebata, or SAP each require a different approach. File exports, REST API, or read replica. Confirm in the first discovery meeting.	D2HIGH Billing Viewer scope (May 2026 go-live) If it's a thin reporting layer, we have ample room. If it's a full billing engine, we must layer above it — not duplicate. Confirm via eTenders Portal procurement notice.	D3HIGH SPV Shareholder Agreement — billing clause Does VCWU bill residents directly, or does Emfuleni Finance bill and remit? This determines whether our customer is Finance only, or also the SPV. Request from CFO office or Rand Water.
D4MEDIUM Water smart meter rollout status Smart meters are currently electricity-focused. Are any water districts already on AMI? If yes, ingest live reads immediately. If no, start with photo reads and estimate correction.	D5MEDIUM Procurement pathway Anything above R200k requires competitive bidding under SA municipal SCM. Determine whether an unsolicited bid (Treasury Reg 16A6.4) or a pilot agreement (lower threshold) is the right entry path.	D6MEDIUM Internal CFO / Revenue sponsor Our contact (former municipal accountant) has offered introductions to the CFO or Directors. This is the single most important non-technical dependency. Take up the offer this week.

How we enter, *get paid, and grow.*

ENTRY STRATEGY

WEEK 1-2	Accept the warm intro offer. Reply to contact — ask who to meet, how, and what he needs in return.
WEEK 3-4	Submit IDP stakeholder comment via public participation channel — free, legitimate, puts us on record.
WEEK 4-6	Deliver concept paper (needs articulation document, not product spec). Figma prototype ready.
MONTH 2-3	First formal meeting. Propose a pilot — small scope, 6 months, clear metrics, low upfront cost.
MONTH 3-6	Execute pilot. Build only what the pilot scope requires. Measure and document outcomes.
MONTH 6+	Contract extension or formal tender response, using pilot results as proof of performance.

REVENUE MODELS

PRIMARY

RECOMMENDED

Outcome-based

Share of incremental collection above an agreed baseline. Aligns incentives with the municipality's foremost priority. Bypasses the 'no budget' objection. SA precedent under MFMA §80.

SECONDARY

Transaction fees

Small fee per payment processed via our gateway. Compounding revenue as adoption grows. Tiered by method — indigent accounts exempt. Replaces what the municipality pays banks.

FUTURE

SaaS licensing

Annual per-account fee once relationship matures. Layer on top of outcome model. Second municipality is far easier than the first — multi-tenant from day one.

What the team owns — *in the next four weeks.*

BUSINESSTHIS WEEK	DEVELOPERSNEXT 2 WEEKS	BOTHWEEKS 3-4
1. Reply to contact — accept the intro, ask who/how/what he needs in return. 2. Also ask: SALGA or CoGTA Gauteng contacts available? 3. Draft one-pager (needs articulation, not product spec). 4. Submit IDP stakeholder comment via public participation channel. 5. Define what 'success' looks like for a 6-month pilot.	1. Confirm billing system vendor via public info or first meeting. 2. Check eTenders Portal — Billing Viewer contract scope. 3. Research USSD aggregators (Clickatell / Flickswitch pricing + API). 4. Spike: Solar / Munsoft integration surface — what does it actually expose? 5. Build Figma prototype: resident view, clerk queue, CFO reconciliation dashboard.	1. Validate one-pager with contact before sending to CFO. 2. Align on commercial model to lead with (outcome-based first). 3. Confirm pilot scope and exact success metrics before the meeting. 4. Agree go / no-go criteria after the first formal meeting. 5. Set a 4-week review date to reassess based on discovery findings.

Emfuleni has named the problem in their own planning document.

We build the revenue-side solution the SPV was never designed to provide.

THE PROBLEM IS REAL

7 qualified audits. 56% NRW. 75% collection. IDP names billing as its foremost priority.

THE TIMING IS URGENT

SPV transition July 2026 adds reconciliation complexity — the window is now.

THE ENTRY IS CLEAN

IDP public participation channel, warm CFO intro, needs-articulation document first.

THE BUILD IS SCOPED

Phase 1 is a dispute tool + resident view — not the whole vision shipped at once.